MATH 189 HW 1

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Question 1

a. Regression, $n = 500$, $p = 3$

b. Categorical, $n = 20$, $p = 14$

c. Regression, $n = 52$, $p = 3$

Question 2

a. Obs. 1: 2
Obs. 2: 2
Obs. 3: $\sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.16$
Obs. 4: $\sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.23$
Obs. 5: $\sqrt{(-1)^2 + 1^2} = \sqrt{2} \approx 1.41$
Obs. 6: $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \approx 1.73$

b. Our prediction for $K = 1$ is Green, because that's the color of the closest neighbor.

c. Our prediction for $K = 3$ is Red, because it is close to two red neighbors and one green neighbors, so by plurality, red is the color of our prediction.

d.